IN4MATX 133: User Interface Software

Lecture 8:
AJAX, Fetch, & Promises

Announcements

- Working on A1 grading, aiming for end of this week
  - My travel delayed things somewhat
  - AWS outage -> Canvas was down
- I owe some of you emails, I’ll get to that later today

Announcements

- You have everything you need in order to complete A2
- This lecture and beyond focuses on A3 content
- A3 will be posted early next week

Today's goals

By the end of today, you should be able to…

- Explain how programs access web resources
  and common ways they respond

- Implement a fetch request to get a resource from a web API

- Use promises to make an asynchronous request

Web APIs

- Many web services and data sources allow you to use HTTP (web) requests to access their data

- This is done by providing a web API.

https://pokeapi.co/

Web APIs

Application Programming Interface

- The *interface* we can use to interact with an *application* through *programming*

- An interface is just a defined set of functions

```
function doSomething(param1, param2) {
//...
}
```

An interface

Web APIs

GitHub

Spotify

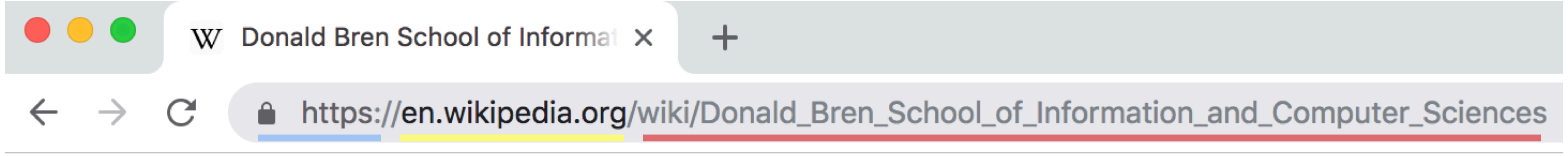
OC Transpo

Google Maps

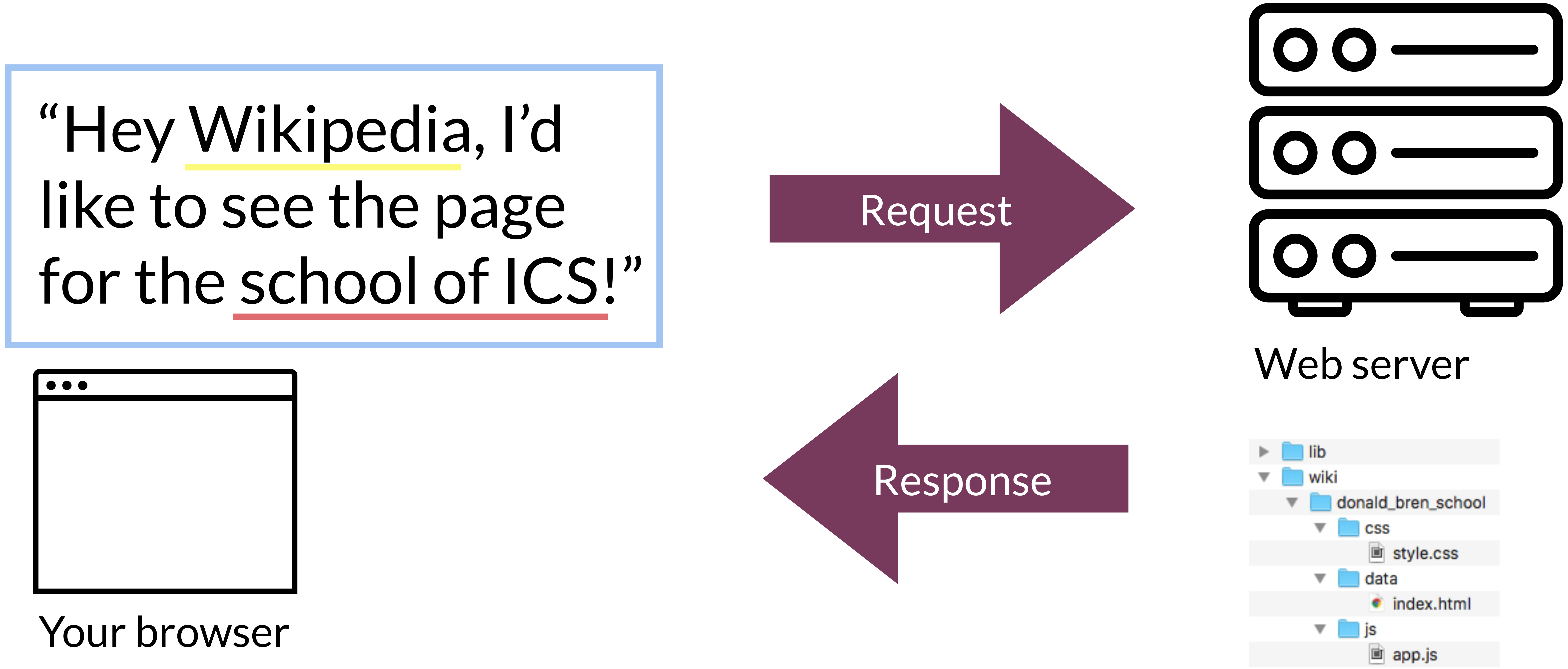
stripe

United States Census Bureau

Using the internet



Protocol (how to handle info) | Host (who has info) | Resource (what info you want)



URI

Uniform Resource Indicator

- All URLs are URIs, but URLs also specify “access mechanism”
  - `http://`, `file://`
- URIs will return a resource
  - Could be a webpage, image file etc.
  - Could also just be data

URI

Uniform Resource Indicator

- `http://www.domain.com/users` => returns a list of users
  - The list of users is the *resource*
- Can have sub-resources
- `http://www.domain.com/users/shawna`
  - Returns a specific user

URI format

- Base URI:
  - How every API request for that API starts
  - `https://pokeapi.co/api/`
- Endpoint
  - Specific resources which can be accessed via that api
  - `v2/pokemon/ditto`
  - `v2/type/3`

Endpoints often contain an API version number

https://pokeapi.co/

URI queries

- Sometimes, APIs include key/value pairs which follow the URI
  - Parameters for the resource, may specify exactly what to return or what format it should be in
  - `?key=value&key=value`
  - Pokeapi doesn't do this, but Spotify does (for A3)
- `https://api.twitter.com/1.1/search/tweets.json?q=UCI&lang=en` language=english

  "query", in Twitter this means
  what text or hashtag to search for

https://developer.twitter.com/en/docs/tweets/search/api-reference/get-search-tweets.html

HTTP verbs

- HTTP requests include a target resource and a verb (method) specifying what to do with it
    - `GET`: return a representation of the current state of the resource
    - `POST`: add a new resource (e.g., a record, an entry)
    - `PUT`: update an existing resource to a new state
    - `PATCH`: update a portion of the resource’s state
    - `DELETE`: remove the resource
    - `OPTIONS`: return a set of methods that can be performed on the resource

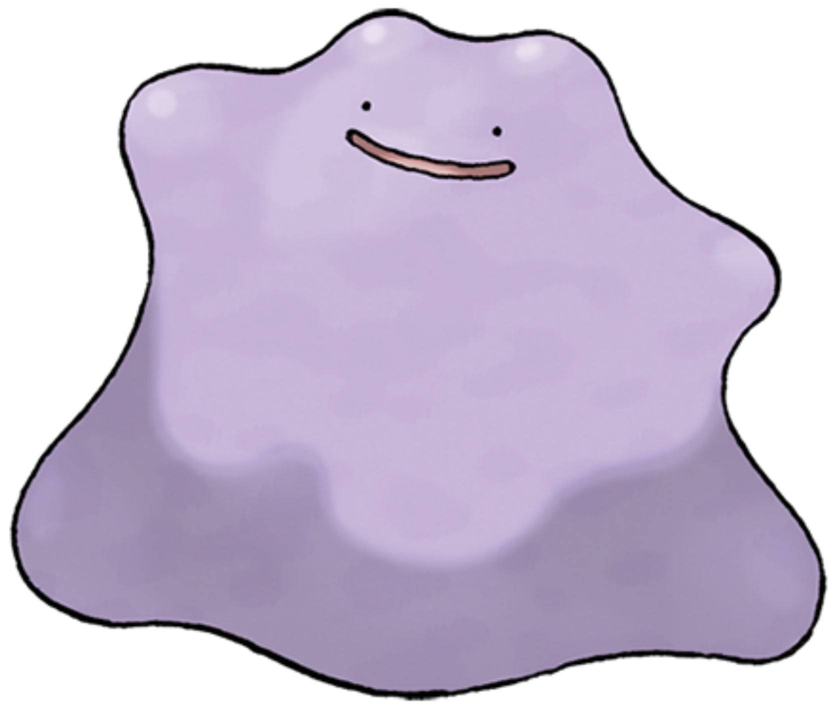
HTTP responses

- Responses will include a *status code* (whether it worked as expected) and a *body* (the actual response)
  - `200:OK`
  - `201:Created` (for `POST`)
  - `400:Bad request` (something is wrong with your URI)
  - `403:Forbidden` (some access or authentication issue)
  - `404:Not found` (resource does not exist)
  - `500:Internal server error` (generic server-side error)

https://www.restapitutorial.com/httpstatuscodes.html

Putting it all together

- `HTTP GET https://pokeapi.co/v2/pokemon/ditto`
  - Use the “get” verb to access the Pokemon “Ditto”
  - We expect/hope for status code `200` (`OK`)
  - Then we access the *body*



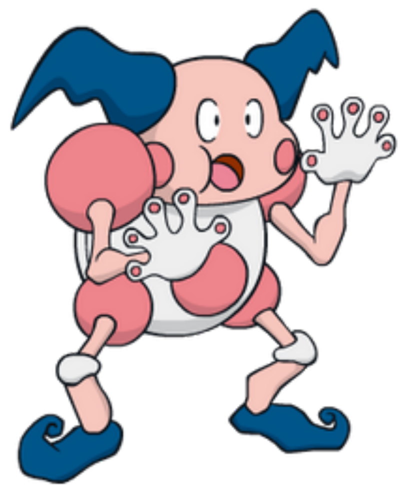
Escaping characters

- Some characters are reserved in URLs, we need to *encode* them
- *Who's that Pokemon? It's Farfetch'd!*
  - We should have to code the apostrophe ('), but instead the API just drops it.
- `HTTP GET https://pokeapi.co/v2/pokemon/farfetchd`

Character	From Windows-1252	From UTF-8
space	%20	%20
!	%21	%21
"	%22	%22
#	%23	%23
$	%24	%24
%	%25	%25
&	%26	%26
'	%27	%27

https://www.w3schools.com/tags/ref_urlencode.asp

Question



Character	From Windows-1252	From UTF-8
space	%20	%20
!	%21	%21
"	%22	%22
#	%23	%23
.	%2E	%2E

If we did need to encode... how would we the Pokeapi for Mr. Mime?

A `HTTP GET https://pokeapi.co/api/v2/pokemon/mr. mime`

B `HTTP GET https://pokeapi.co/api/v2/pokemon/mrmime`

C `HTTP GET https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`

D `HTTP POST https://pokeapi.co/api/v2/pokemon/mr. mime`

E `HTTP POST https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`

Question

Character	From Windows-1252	From UTF-8
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If we did need to encode... how would we the Pokeapi for Mr. Mime?

- (A) `HTTP GET https://pokeapi.co/api/v2/pokemon/mr. mime`
- (B) `HTTP GET https://pokeapi.co/api/v2/pokemon/mrmime`
- (C) `HTTP GET https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`
- (D) `HTTP POST https://pokeapi.co/api/v2/pokemon/mr. mime`
- (E) `HTTP POST https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`

A 0%

B 0%

C 0%

D 0%

E 0%

Question

Character	From Windows-1252	From UTF-8
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C `HTTP GET https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`

D `HTTP POST https://pokeapi.co/api/v2/pokemon/mr. mime`

E `HTTP POST https://pokeapi.co/api/v2/pokemon/mr%2E%20mime`

So how do we make a web request?

Asynchronous JavaScript and XML

XML

Extensible Markup Language

- A generalized syntax for semantically defining structured content
- HTML is XML with defined tags

```
<person>
   <firstName>Alice</firstName>
   <lastName>Smith</lastName>
   <favorites>
      <music>jazz</music>
      <food>pizza</food>
   </favorites>
</person>
```

Plain text

```
Belgian Waffles
"Two of our famous Belgian Waffles with plenty of real maple syrup"
$5.95
650 calories


Strawberry Belgian Waffles
"Light Belgian waffles covered with strawberries and whipped cream"
$7.95
900 calories


Berry-Berry Belgian Waffles
"Light Belgian waffles covered with an assortment of fresh berries and whipped
cream"
$8.95
900 calories


French Toast
"Thick slices made from our homemade sourdough bread"
$4.50
600 calories


Homestyle Breakfast
"Two eggs, bacon or sausage, toast, and our ever-popular hash browns"
$6.95
950 calories
```

XML

```xml
<breakfast_menu>
  <food>
    <name>Belgian Waffles</name>
    <price>$5.95</price>
    <description>
      Two of our famous Belgian Waffles with plenty of real maple syrup
    </description>
    <calories>650</calories>
  </food>
  <food>
    <name>Strawberry Belgian Waffles</name>
    <price>$7.95</price>
    <description>
      Light Belgian waffles covered with strawberries and whipped cream
    </description>
    <calories>900</calories>
  </food>
  <food>
    <name>Berry-Berry Belgian Waffles</name>
    <price>$8.95</price>
    <description>
      Light Belgian waffles covered with an assortment of fresh berries and whipped
cream
    </description>
    <calories>900</calories>
  </food>
  <food>
    <name>French Toast</name>
    <price>$4.50</price>
    <description>
      Thick slices made from our homemade sourdough bread
    </description>
    <calories>600</calories>
  </food>
  <food>
    <name>Homestyle Breakfast</name>
    <price>$6.95</price>
    <description>
      Two eggs, bacon or sausage, toast, and our ever-popular hash browns
    </description>
    <calories>950</calories>
  </food>
</breakfast_menu>
```

XML

```xml
<breakfast_menu>
  <food>
    <name>Belgian Waffles</name>
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    <description>
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  </food>
  <food>
    <name>Homestyle Breakfast</name>
    <price>$6.95</price>
    <description>
      Two eggs, bacon or sausage, toast, and our ever-popular hash browns
    </description>
    <calories>950</calories>
  </food>
</breakfast_menu>
```

JSON

```json
{
  "breakfast_menu": {
    "food": [
    {
      "name": "Belgian Waffles",
      "price": "$5.95",
      "description": "Two of our famous Belgian Waffles with plenty of real maple
syrup",
      "calories": "650"
    },
    {
      "name": "Strawberry Belgian Waffles",
      "price": "$7.95",
      "description": "Light Belgian waffles covered with strawberries and whipped
cream",
      "calories": "900"
    },
    {
      "name": "Berry-Berry Belgian Waffles",
      "price": "$8.95",
      "description": "Light Belgian waffles covered with an assortment of fresh
berries and whipped cream",
      "calories": "900"
    },
    {
      "name": "French Toast",
      "price": "$4.50",
      "description": "Thick slices made from our homemade sourdough bread",
      "calories": "600"
    },
    {
      "name": "Homestyle Breakfast",
      "price": "$6.95",
      "description": "Two eggs, bacon or sausage, toast, and our ever-popular hash
browns",
      "calories": "950"
    }
    ]
  }
}
```

XML vs. JSON

- XML and JSON represent the same data
- JSON is more concise
  - Less data to move around on the web
- JSON is easier to read
  - Close tags in XML are redundant
- JSON has taken over as the typical format of web requests

AJAX

Asynchronous JavaScript and ~~XML~~

JSON

Sending an AJAX request

XMLHttpRequest

- AJAX requests are built into a browser-provided object called `XMLHTTPRequest`

```
var xhttp = new XMLHttpRequest();
xhttp.onreadystatechange = function() {
    if (xhttp.readyState == 4 && xhttp.status == 200) {
        // Action to be performed when the document is read;
        var xml = xhttp.responseXML;

        var movie = xml.getElementsByTagName("track");
        //...
    }
};
xhttp.open("GET", "filename", true);
xhttp.send();
```

XMLHttpRequest

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       var xml = xhttp.responseXML;

       var movie = xml.getElementsByTagName("track");
       //...
    }
};
xhttp.open("GET", "filename", true);
xhttp.send();
```

Fetch

- A new, modern method for submitting XMLHttpRequests
- Included in most browsers
- `fetch('url')`

Chrome	Edge *	Safari	Firefox	Opera	IE	Chrome for Android	Safari on iOS *
4-39			2-33	10-26			
[2] 40			[1 4] 34-38	[2] 27			
[2 3] 41	12-13	3.1-10	[4] 39	[2 3] 28			3.2-10.2
42-140	14-140	10.1-18.6	40-142	29-121	6-10		10.3-18.6
141	141	26.0	143	122	11	141	26.0
142-144		26.1-TP	144-146				26.1

Using fetch

- `fetch(‘some-url’)` defaults to a `GET` request
- `fetch` can optionally take a second `options` argument (as a dictionary)
  - `method`: what method to use (e.g., `POST`, `PUT`, `DELETE`)
  - `headers`: specify content type format, etc. (more on headers in the next week)
  - `body`: what you want to send for a `POST`/`PUT` request

https://css-tricks.com/using-fetch/

Using fetch

- For a `GET` request

```
fetch('some-url');
```

- For a `POST` request

```
fetch('some-url', {
 method:'POST',
 headers: {'Content-Type': 'application/json'},
 body: JSON.stringify(data-to-send)
});
```

https://css-tricks.com/using-fetch/

A local web server

- Install live-server package globally
  - `npm install -g live-server`
- Running it
  - `cd path/to/project`
  - `live-server .`
- Will open up your webpage at http://localhost:8080

Asynchronous JavaScript and ~~XML~~

JSON

Asynchronous requests

- Ajax requests are asynchronous, so they happen simultaneously with the rest of the code
- After the request is sent, the next line of code is executed **without waiting for the request to finish**

(1)
```
console.log('About to send request');

//send request for data to the url
```
(2)
```
fetch(url);
```
Does **NOT** return the data

(3)
```
console.log('Sent request’);
```

(4) Data is actually received sometime later!

Asynchronous requests

- It’s uncertain how long it’ll take the request to complete
- Handling requests asynchronously allows a person to continue interacting with your page
    - The request is not blocking their interface interactions
    - It’s a bad experience when a person tries to navigate your webpage, but can’t

Promises

- Because `fetch()` is asynchronous, the method returns a **Promise**

- Promises act as a "placeholder" for the data
  that will eventually be received from the AJAX request

```
//fetch() returns a Promise
var thePromise = fetch(url);
```

Promises

- We use the `.then()` method to specify a **callback** function to be executed when the promise is *fulfilled* (when the asynchronous request is finished)

```
//what to do when we get the response
function successCallback(response) {
   console.log(response);
}
```

Callback will be passed the request response

```
//when fulfilled, execute the callback function
//(which will be passed the fetched data)
var promise = fetch(url);
promise.then(successCallback, rejectCallback);
```

Optional parameter

```
//more common to use anonymous variables/callbacks:
fetch(url).then(function(response) {
   console.log(response);
});
```

fetch() responses

- The parameter passed to the `.then()` callback is the **response**, not the data we're looking for
- The fetch() API provides a method `.json()` that we can use to extract the data from the response
    - But this method is *also* asynchronous and returns a promise!

```
fetch(url).then(function(response) {
   var newPromise = response.json();
```

Not the data

Another promise

```
   //... what now?
});
```

Chaining promises

- The `.then()` method itself returns a Promise containing the value (data) returned by the callback method
- This allows you to **chain** callback functions together,
  doing one after another (but *after* the Promise is fulfilled)

```
function makeString(data) {
   return data.join(", "); //a value to put in Promise
}

function makeUpper(string) {
   return string.toUpperCase(); //a value to put in Promise
}

var promiseA = getData();
var promiseB = promiseA.then(makeString);
var promiseC = promiseB.then(makeUpper);
promiseC.then(function(data) {
   console.log(data);
};
```

When completed, promiseA => json data
promiseB => comma-separated string
promiseC => uppercase string

Data is an uppercase,
comma-separated string

Chaining promises

- The `.then()` method itself returns a Promise containing the value (data) returned by the callback method

- This allows you to **chain** callback functions together,
  doing one after another (but *after* the Promise is fulfilled)

```
function makeString(data) {
   return data.join(", "); //a value to put in Promise
}

function makeUpper(string) {
   return string.toUpperCase(); //a value to put in Promise
}

//more common to use anonymous variables and chain functions
getData()
   .then(makeString)
   .then(makeUpper)
   .then(function(d) { console.log(d); };
```

Multiple promises (sequential)

- The .then() function will also handle promises *returned by previous callbacks*, allowing for sequential async calls

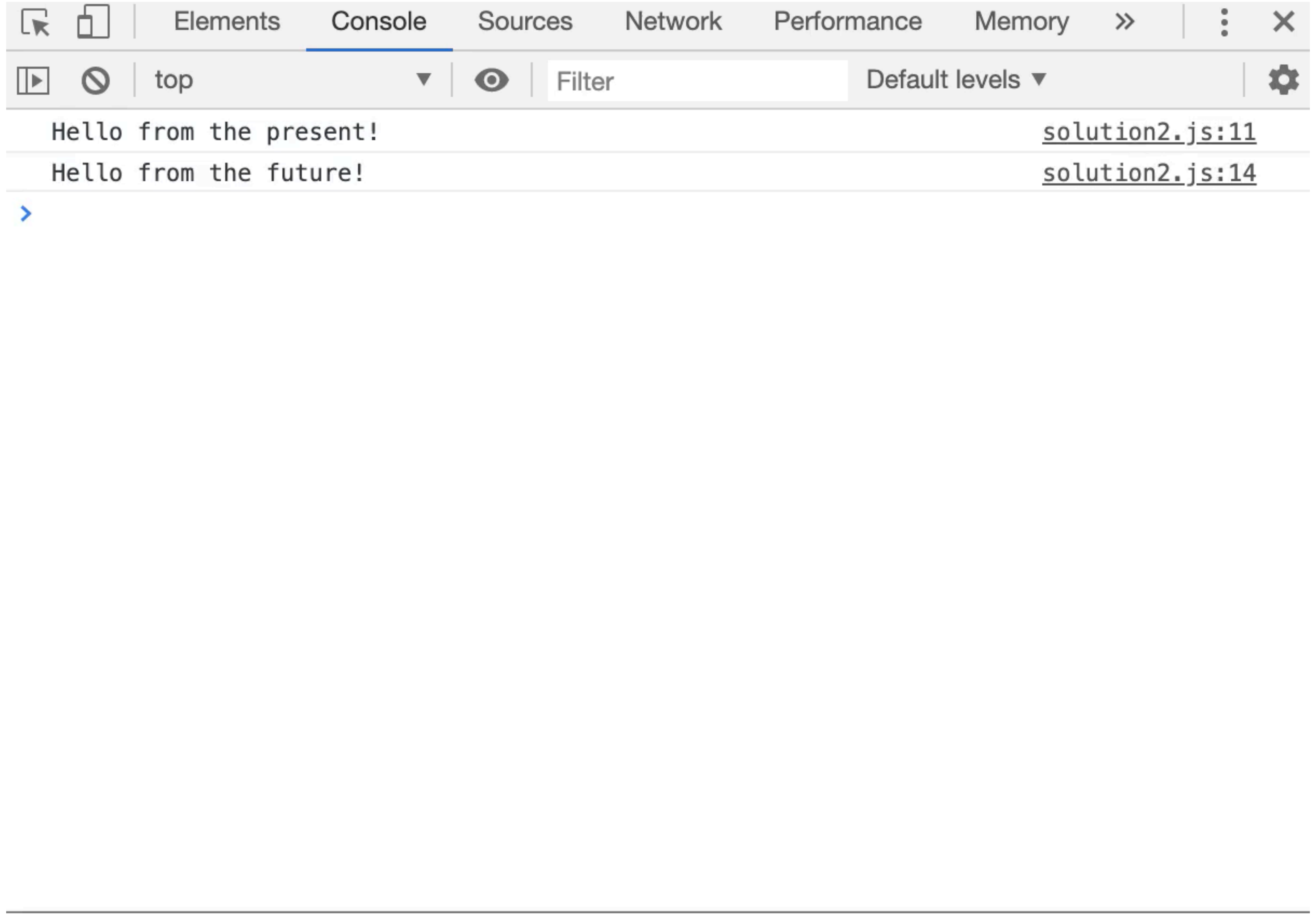
```
getData(fooSrc)
  .then(function(fooData){
    var modifiedFoo = modify(fooData)
    return modifiedFoo;
  })
  .then(function(modifiedFoo){
    //do something with modifiedFoo
    var barPromise = getData(barSrc);
    return barPromise;
  })
  .then(function(barData){
    //do something with barData
  })
```

Extracting fetch() data

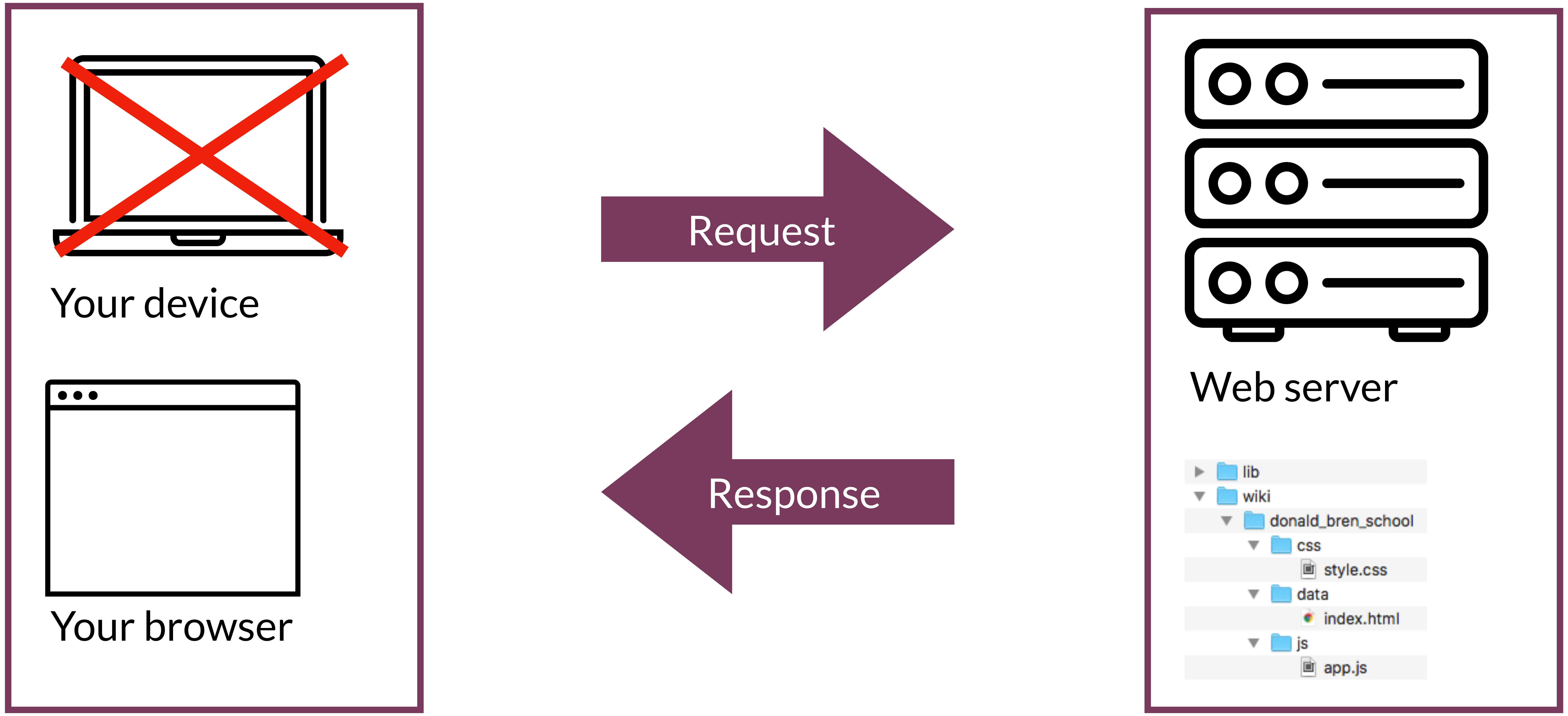
- To actually download JSON data...

```
fetch(url)
    .then(function(response) {
        var dataPromise = response.json();
        return dataPromise;
    })
    .then(function(data) {
        //do something with data
    });
```

Promises


Elements
Console
Sources
Network
Performance
Memory
top
Filter
Default levels
Hello from the present!
solution2.js:11
Hello from the future!
solution2.js:14

Client-side and server-side JavaScript

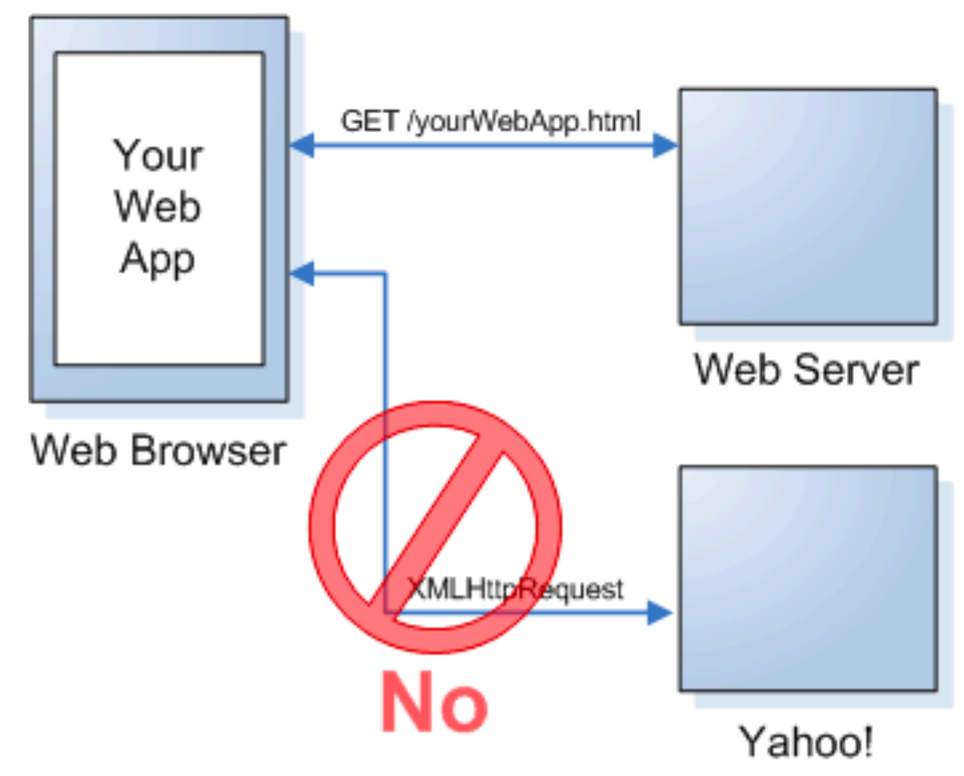


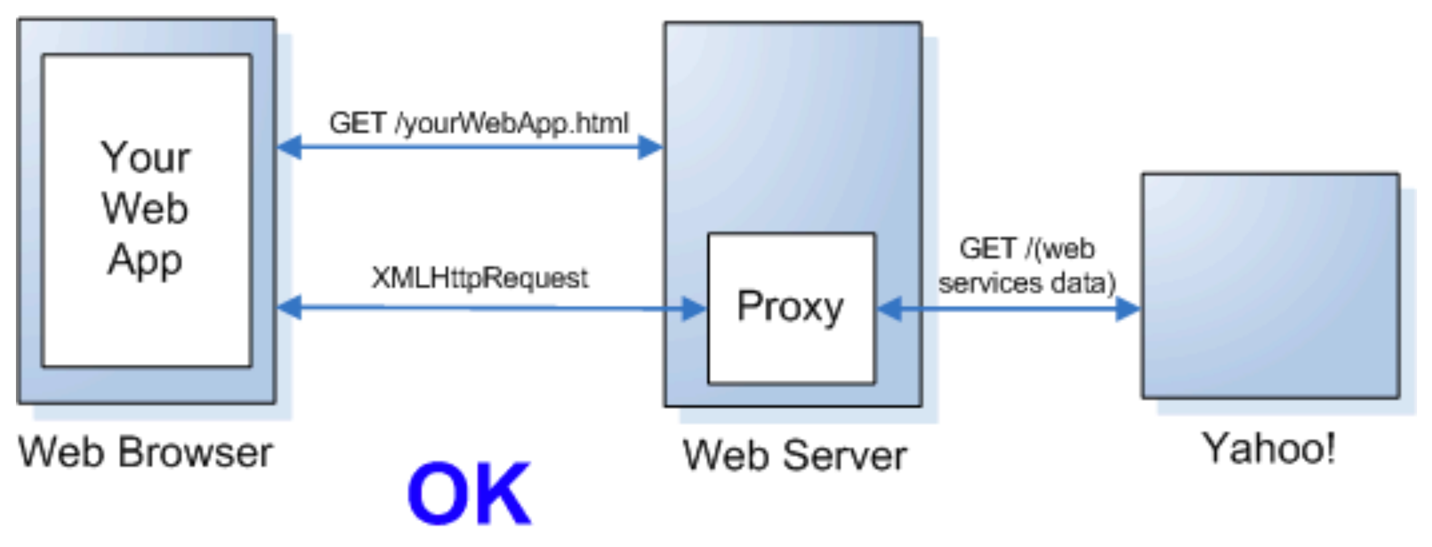
Edit what's being rendered
Trigger or react to events

Navigate file system programmatically
Dynamically generate pages or views
Transport, store, or interact with data

Same-origin policy

- Many browsers will not permit AJAX requests to a different server.
  This helps prevent malicious scripts from accessing data in the DOM
  - A non-browser proxy server running locally can communicate with a different server
  - The browser can communicate with the proxy server





https://en.wikipedia.org/wiki/Same-origin_policy

Same-origin policy

- Two browser tabs: a bank app is open in one, an evil app in the other
  - Both have JavaScript scripts
  - The bank uses it's script to read, edit, etc. your bank data
- The *origin* is what HTML page opened the JavaScript file
  - So each tab is a separate origin
- *Without* the same-origin policy, the evil app could read, edit, etc. your bank data
  - Different tabs, but both running with the same JavaScript engine

https://security.stackexchange.com/questions/8264/why-is-the-same-origin-policy-so-important

Same-origin policy

- So instead, the bank's JavaScript script can only perform actions in it's browser tab (origin), and the evil app's JavaScript script can only perform actions in it's browser tab

- Two exceptions:
  - An app can always communicate with other apps in the same domain (e.g., localhost apps can communicate with any other localhost apps*)
  - A server can designate that it will accept connections from sources with a particular origin (or any origin)
  - You *can* disable the policy in your browser, but probably shouldn't

https://security.stackexchange.com/questions/8264/why-is-the-same-origin-policy-so-important

Client-side

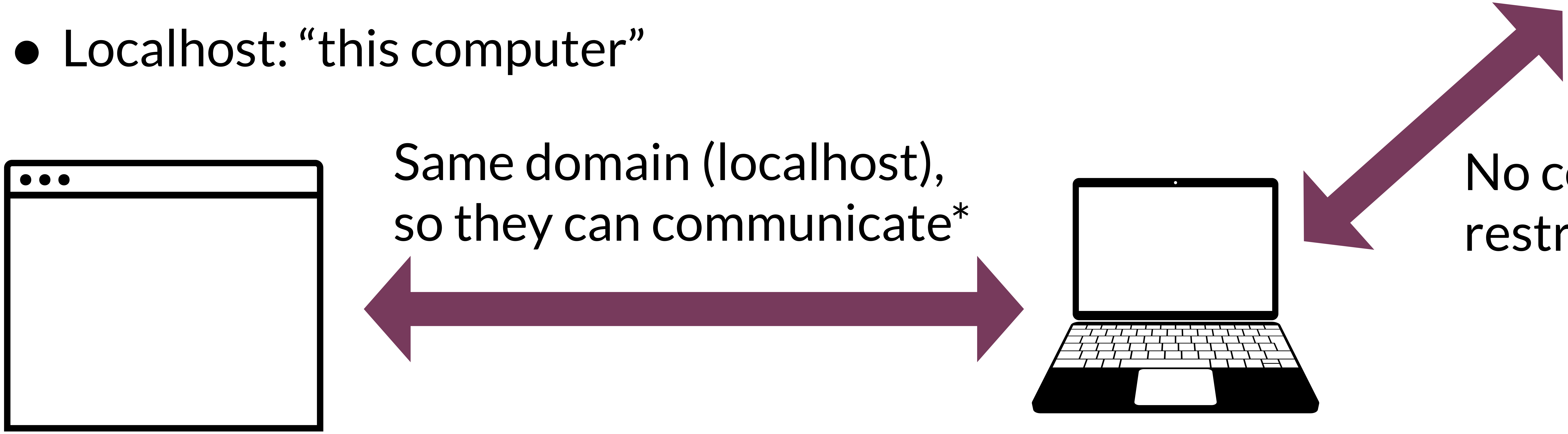
- Runs in the browser
- Changes happen in real-time in the browser
- Cannot make HTTP requests to many APIs
- Examples: AJAX, Angular, React, Vue.js

Server-side

- Runs in the command line, etc. (but maybe can still be accessed from the browser)
- Changes happen in response to HTTP requests
- Can make HTTP requests to most APIs
- Examples: Node, ASP.NET

Servers on localhost

- Localhost: “this computer”



Live server: localhost:8080

Browser implements same-origin policy to protect the other data you have open in the browser

Spotify proxy: localhost:7890

Running as a server, so no same-origin policy restrictions. Can communicate with Spotify

Question

Which can make an HTTP request to the Spotify API?

(Assume the browser uses default settings)

- (A) 4
- (B) 1, 4
- (C) 1, 2, 4
- (D) 1, 3, 4
- (E) 1, 2, 3, 4

(1) A browser open to spotify.com

(2) A browser with client-side JavaScript at localhost:8888

(3) A browser with server-side JavaScript at localhost:8888

(4) A server running in the Spotify domain

Which can make an HTTP request to the Spotify API?

(Assume the browser uses default settings)

Ⓐ 4

Ⓑ 1, 4

Ⓒ 1, 2, 4

Ⓓ 1, 3, 4

Ⓔ 1, 2, 3, 4

(1) A browser open to spotify.com

(2) A browser with client-side JavaScript at localhost:8888

(3) A browser with server-side JavaScript at localhost:8888

(4) A server running in the Spotify domain

A		0%
B		0%
C		0%
D		0%
E		0%

Question

Which can make an HTTP request to the Spotify API?

(Assume the browser uses default settings)

A 4

B 1, 4

C 1, 2, 4

D 1, 3, 4

E 1, 2, 3, 4

(1) A browser open to spotify.com

(2) A browser with client-side JavaScript at localhost:8888

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(4) A server running in the Spotify domain

Getting pokemon

First 25 pokemon

- #1: bulbasaur

- #2: ivysaur

- #3: venusaur

- #4: charmander

- #5: charmeleon

- #6: charizard

- #7: squirtle

Question

What will (probably) be shown in the console?

```
favorite_pet = "dogs";

var url = 'https://fakeapi.com/send_cats';
fetch(url).then((response) => {
  //sends back the string "cats"
  console.log("I like " + favorite_pet);
  response.json().then((pet) => {
    favorite_pet = pet;
    console.log("I like " + favorite_pet);
  });
});

console.log("I like " + favorite_pet);
```

A
```
I like dogs
I like cats
I like dogs
```

B
```
I like dogs
I like dogs
I like dogs
```

C
```
I like cats
I like dogs
I like dogs
```

D
```
I like dogs
I like dogs
I like cats
```

E
```
I like dogs
I like cats
I like cats
```

```
favorite_pet = "dogs";

var url = 'https://fakeapi.com/send_cats';
fetch(url).then((response) => {
  //sends back the string "cats"
  console.log("I like " + favorite_pet);
  response.json().then((pet) => {
    favorite_pet = pet;
    console.log("I like " + favorite_pet);
  });
});

console.log("I like " + favorite_pet);
```

A
```
I like dogs
I like cats
I like dogs
```

B
```
I like dogs
I like dogs
I like dogs
```

C
```
I like cats
I like dogs
I like dogs
```

D
```
I like dogs
I like dogs
I like cats
```

E
```
I like dogs
I like cats
I like cats
```

A 0%

B 0%

C 0%

D 0%

E 0%

Question

What will (probably) be shown in the console?

```
favorite_pet = "dogs";

var url = 'https://fakeapi.com/send_cats';
fetch(url).then((response) => {
  //sends back the string "cats"
  console.log("I like " + favorite_pet);
  response.json().then((pet) => {
    favorite_pet = pet;
    console.log("I like " + favorite_pet);
  });
});

console.log("I like " + favorite_pet);
```

A
```
I like dogs
I like cats
I like dogs
```

B
```
I like dogs
I like dogs
I like dogs
```

C
```
I like cats
I like dogs
I like dogs
```

D
```
I like dogs
I like dogs
I like cats
```

E
```
I like dogs
I like cats
I like cats
```

Catching errors

- We can use the `.catch()` function to specify a callback that will occur if the promise is **rejected** (an error occurs).
  - This method will "catch" errors from all previous `.then()`s

```
getData(fooSrc)
   .then(firstCallback)
   .then(secondCallback)
   .catch(function(error) {
      //called if EITHER previous callback
      //has an error

      //param is object representing the error itself
      console.log(error.message);
   })
   .then(thirdCallback) //will only do this if
                        //no previous errors
```

Multiple promises (concurrent)

- Because Promises are just commands to do something, we can wait for all of them to be done

```
var foo = fetch(fooUrl);
var bar = fetch(barUrl);

//a promise for when all commands ready
Promise.all(foo, bar)
  .then(function(fooRes, barRes) {
     //do something both both responses, e.g.,

     return Promise.all(fooRes.json(), barRes.json());

  })
  .then(function(fooData, barData){
     //now have both data sets!
  })
```

Today's goals

By the end of today, you should be able to...

- Explain how programs access web resources and common ways they respond
- Implement a fetch request to get a resource from a web API
- Use promises to make an asynchronous request

IN4MATX 133: User Interface Software

Lecture 8:
AJAX, Fetch, & Promises